

Chiral relations for entanglement in quantum self phase modulation

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An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

I. INTRODUCTION

II. SPATIALLY STRUCTURED BOSONIC FIELD

Consider a bosonic quantum field $\mathcal{E}(\mathbf{r})$ satisfying the commutation relations

$$[\mathcal{E}(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)] = \delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2), \quad [\mathcal{E}(\mathbf{r}_1), \mathcal{E}(\mathbf{r}_2)] = [\mathcal{E}^\dagger(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)] = 0 \quad (1)$$

In the above equation and in what follows, $\mathbf{r} = (x, y)$ is the transverse coordinate.

One can introduce ladder operators $a_v, a_v^\dagger$ associated with the spatial mode v by the expression

$$a(v) = \int_{\mathbb{R}^2} v^*(\mathbf{r}) \mathcal{E}(\mathbf{r}) d\mathbf{r}. \quad (2)$$

We then have

$$[a(v_1), a(v_2)^\dagger] = \int_{\mathbb{R}^2} v_1^*(\mathbf{r}) v_2(\mathbf{r}) d\mathbf{r} = \langle v_1 | v_2 \rangle, \quad (3)$$

while $[a(v_1), a(v_2)] = [a(v_1)^\dagger, a(v_2)^\dagger] = 0$.

Let us define the quadrature associated to the spatial mode v as

$$\begin{aligned} X(v) &= \frac{1}{\sqrt{2}} \int_{\mathbb{R}^2} v^*(\mathbf{r}) \mathcal{E}(\mathbf{r}) + v(\mathbf{r}) \mathcal{E}^\dagger(\mathbf{r}) d\mathbf{r} \\ &= \frac{a(v) + a(v)^\dagger}{\sqrt{2}} \end{aligned} \quad (4)$$

We then have the commutation relation

$$[X(v_1), X(v_2)] = i \operatorname{Im} \langle v_1 | v_2 \rangle, \quad (5)$$

which leads to the uncertainty principle

$$\Delta X(v_1)^2 \Delta X(v_2)^2 \geq \frac{|\operatorname{Im} \langle v_1 | v_2 \rangle|^2}{4}. \quad (6)$$

One could also introduce the complementary quadrature $P(v) = X(-iv)$ so that

$$[X_{v_1}, P_{v_2}] = i \operatorname{Re} \langle v_1 | v_2 \rangle \quad (7)$$

and

$$\Delta X(v_1)^2 \Delta P(v_2)^2 \geq \frac{|\operatorname{Re} \langle v_1 | v_2 \rangle|^2}{4}. \quad (8)$$

We are interested in observables of the form

$$\Gamma(v_1, v_2) = \langle \{ \Delta X(v_1), \Delta X(v_2) \} \rangle = \operatorname{Re} [\langle v_1 | v_2 \rangle + 2f(v_1, v_2) + 2h(v_1, v_2)], \quad (9)$$

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where we have defined

$$\begin{aligned} f(v_1, v_2) &= \langle a(v_1)a(v_2) \rangle - \langle a(v_1) \rangle \langle a(v_2) \rangle \\ h(v_1, v_2) &= \langle a(v_2)^\dagger a(v_1) \rangle - \langle a(v_2)^\dagger \rangle \langle a(v_1) \rangle \end{aligned} \quad (10)$$

In terms of the fields, the required quantities are written as

$$f(v_1, v_2) = \int v_1^*(\mathbf{r}_2) v_2^*(\mathbf{r}_1) F(\mathbf{r}_1, \mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2; \quad F(\mathbf{r}_1, \mathbf{r}_2) = \langle \mathcal{E}(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \rangle - \langle \mathcal{E}(\mathbf{r}_1) \rangle \langle \mathcal{E}(\mathbf{r}_2) \rangle \quad (11)$$

$$h(v_1, v_2) = \int v_1^*(\mathbf{r}_1) v_2(\mathbf{r}_2) H(\mathbf{r}_1, \mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2; \quad H(\mathbf{r}_1, \mathbf{r}_2) = \langle \mathcal{E}^\dagger(\mathbf{r}_2, z) \mathcal{E}(\mathbf{r}_1, z) \rangle - \langle \mathcal{E}^\dagger(\mathbf{r}_2, z) \rangle \langle \mathcal{E}(\mathbf{r}_1, z) \rangle \quad (12)$$

This formulation is useful because f and g are normally ordered expectation values, which can be more easily calculated. Furthermore, this representation also makes it easier to perform linear combinations of the modes, which will be useful later. These combinations will be based on the properties

$$\begin{aligned} f(v_1, v_2 + \lambda v_3) &= f(v_1, v_2) + \lambda^* f(v_1, v_3) \\ f(v_1, v_2) &= f(v_2, v_1) \\ h(v_1, v_2 + \lambda v_3) &= h(v_1, v_2) + \lambda h(v_1, v_3) \\ h(v_1, v_2) &= h(v_2, v_1)^* \end{aligned} \quad (13)$$

which are a reflection of the fact that $a(v)$ is antilinear while $a^\dagger(v)$ is linear.

A. One mode correlation

Let us analyze the correlation matrix $\gamma_{jk} = \langle \{Z_j, Z_k\} \rangle - 2 \langle Z_j \rangle \langle Z_k \rangle$ where $\mathbf{Z} = (X(v), P(v))$. We observe that $h(v, v) \in \mathbb{R}$, so we get

$$\gamma = \begin{pmatrix} 1 + 2h(v, v) & 0 \\ 0 & 1 + 2h(v, v) \end{pmatrix} + 2 \begin{pmatrix} \text{Re } f(v, v) & -\text{Im } f(v, v) \\ -\text{Im } f(v, v) & -\text{Re } f(v, v) \end{pmatrix}. \quad (14)$$

Notice that $f(e^{i\phi}v, e^{i\phi}v) = e^{-i2\phi}f(v, v)$. Therefore, we can diagonalize γ by choosing $\phi = [\arg f(v, v) \pm \pi]/2$. We then get eigenvalues

$$\lambda_{\pm} = 1 + 2h(v, v) \pm 2|f(v, v)|. \quad (15)$$

We then see that the quadrature rotated along the axis ϕ has minimum variance of λ_- , while the one rotated along the axis $\phi + \pi/2$ has maximum variance λ_+ . Furthermore, we have squeezing if $|f(v, v)| > h(v, v)$.

B. Duan Criterion

Consider two orthonormal states v_1, v_2 , i.e., $\langle v_j | v_k \rangle = \delta_{jk}$. The Duan criterion [1] states that if

$$D = \frac{1}{2} \langle [\Delta X(v_1) + \Delta X(v_2)]^2 \rangle + \frac{1}{2} \langle [\Delta P(v_1) - \Delta P(v_2)]^2 \rangle < 1, \quad (16)$$

then the states are entangled.

Let us define $v_{\pm} = (v_1 \pm v_2)/\sqrt{2}$. We then have

$$D = \langle \Delta X(v_+)^2 \rangle + \langle \Delta P(v_-)^2 \rangle. \quad (17)$$

Applying (9) we obtain

$$\begin{aligned} D &= 1 + h(v_+, v_+) + h(v_-, v_-) + \text{Re}[f(v_+, v_+) - f(v_-, v_-)] \\ &= 1 + h(v_1, v_1) + h(v_2, v_2) + 2\text{Re } f(v_1, v_2) \end{aligned} \quad (18)$$

Once again, we can maximize the possible violation by rotating the quadrature of the modes, say, v_1 by an angle $\phi = [\arg f(v_1, v_2) \pm \pi]/2$. We get an optimal possible violation of

$$D_{\text{opt}} = 1 + h(v_1, v_1) + h(v_2, v_2) - 2|f(v_1, v_2)| \quad (19)$$

III. THE KERR SYSTEM

A. Heisenberg Equation for the Field and Linear Solution

We will assume that the field is propagating along the z direction, and satisfies the Heisenberg equation of motion

$$2ik\partial_z\mathcal{E}(\mathbf{r}, z) + \nabla^2\mathcal{E}(\mathbf{r}, z) = g\mathcal{E}^\dagger(\mathbf{r}, z)\mathcal{E}^2(\mathbf{r}, z). \quad (20)$$

Let us decompose $\mathcal{E} = \mathcal{E}_0 + \delta\mathcal{E}$ where $\mathcal{E}_0$ is the solution of

$$2ik\partial_z\mathcal{E}_0(\mathbf{r}, z) + \nabla^2\mathcal{E}_0(\mathbf{r}, z) = 0, \quad \mathcal{E}_0(\mathbf{r}, 0) = \mathcal{E}(\mathbf{r}, 0). \quad (21)$$

Let u be a solution of the paraxial wave equation $2ik\partial_z u(\mathbf{r}, z) + \nabla^2 u(\mathbf{r}, z) = 0$. Then, the annihilation operator corresponding to mode u is given by

$$a(u, z) = \int u^*(\mathbf{r}, z)\mathcal{E}(\mathbf{r}, z)d\mathbf{r} = a_0(u) + \delta a(u, z) \quad (22)$$

where

$$a_0(u) = \int u^*(\mathbf{r}, z)\mathcal{E}_0(\mathbf{r}, z)d\mathbf{r} \quad (23)$$

is independent of z (take a z derivative, use the equations of motion and integrate by parts) and

$$\delta a(u, z) = \int u^*(\mathbf{r}, z)\delta\mathcal{E}(\mathbf{r}, z)d\mathbf{r} \quad (24)$$

Our initial state will be a coherent one:

$$|\alpha\rangle = D(\alpha)|0\rangle = \exp(a(\alpha)^\dagger - a(\alpha))|0\rangle. \quad (25)$$

Here, $\alpha(\mathbf{r})$ is a non normalized spatial function, with norm $\|\alpha\|$. This state is an eigenvector of the free field:

$$\mathcal{E}_0(\mathbf{r}, z)|\alpha\rangle = \alpha(\mathbf{r}, z)|\alpha, u\rangle \quad (26)$$

The variation $\delta\mathcal{E}$ satisfies

$$2ik\partial_z\delta\mathcal{E}(\mathbf{r}, z) + \nabla^2\delta\mathcal{E}(\mathbf{r}, z) = g\mathcal{E}^\dagger(\mathbf{r}, z)\mathcal{E}^2(\mathbf{r}, z). \quad (27)$$

Evaluating this at $z = 0$ gives us

$$2ik\partial_z\delta\mathcal{E}(\mathbf{r}, 0) = g\mathcal{E}_0^\dagger(\mathbf{r}, 0)\mathcal{E}_0^2(\mathbf{r}, 0) \quad (28)$$

from which we obtain the Taylor expansion

$$\delta\mathcal{E}(\mathbf{r}, z) \approx -\frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r}, 0)\mathcal{E}_0^2(\mathbf{r}, 0) \quad (29)$$

For the complete field we then have

$$\mathcal{E}(\mathbf{r}, z) \approx \mathcal{E}_0(\mathbf{r}, 0) - \frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r}, 0)\mathcal{E}_0^2(\mathbf{r}, 0) \quad (30)$$

This expression will be the basis of our further developments.

B. Expectation value of field

Taking expectation values, we arrive at

$$\langle\mathcal{E}(\mathbf{r}, z)\rangle \approx \alpha(\mathbf{r}, 0) - \frac{igz}{2k}|\alpha(\mathbf{r}, 0)|^2\alpha(\mathbf{r}, 0). \quad (31)$$

The expectation value of the annihilation operator a_v corresponding to a spatial mode v is then

$$\langle a_v \rangle = \langle v | \alpha \rangle - \frac{igz}{2k} \langle v, \alpha | \alpha, \alpha \rangle \quad (32)$$

In the above expression we have introduced the second spatial overlap

$$\langle v_1, v_2 | u_1, u_2 \rangle = \int v_1^*(\mathbf{r}, z) v_2^*(\mathbf{r}, z) u_1(\mathbf{r}, z) u_2(\mathbf{r}, z) d\mathbf{r}. \quad (33)$$

When $\alpha = \|\alpha\| u_{0l}$ is a Laguerre-Gaussian mode with radial order $p = 0$ and topological charge l , we have that

$$|u_{0l}(\mathbf{r}, z)| u_{0l}^2(\mathbf{r}, z) = \sum_{q=0}^{|l|} C_{ql} \bar{u}_{ql}(\mathbf{r}, z) \quad (34)$$

where $\bar{u}_{ql}$ are Laguerre-Gaussian modes with a reduced waist. Therefore, one may write

$$\langle a_v \rangle = \|\alpha\| \langle v | u_{0,l} \rangle - \frac{igz \|\alpha\|^3}{2k} \sum_{q=0}^{|l|} C_{ql} \langle v | \bar{u}_{ql} \rangle \quad (35)$$

In particular, when $v = \bar{u}_{ql}$, we have that

$$\langle a_{\bar{u}_{ql}} \rangle = \begin{cases} \|\alpha\| \langle \bar{u}_{ql} | u_{0,l} \rangle - \frac{igz \|\alpha\|^3 C_{ql}}{2k} & q \leq |l| \\ \|\alpha\| \langle \bar{u}_{ql} | u_{0,l} \rangle & q > |l| \end{cases}. \quad (36)$$

C. Correlation functions

One of the two point field function is

$$\mathcal{E}^\dagger(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z) \delta \mathcal{E}(\mathbf{r}_2, z) + \delta \mathcal{E}^\dagger(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) \quad (37)$$

where we have only kept terms linear in $\delta \mathcal{E}$.

Close to $z = 0$, we have that

$$\mathcal{E}^\dagger(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1, 0) \mathcal{E}_0(\mathbf{r}_2, 0) - \frac{igz}{2k} \left[\mathcal{E}_0^\dagger(\mathbf{r}_1, 0) \mathcal{E}_0^\dagger(\mathbf{r}_2, 0) \mathcal{E}_0^2(\mathbf{r}_2, 0) - \mathcal{E}_0^\dagger(\mathbf{r}_1, 0)^2 \mathcal{E}_0(\mathbf{r}_1, 0) \mathcal{E}_0(\mathbf{r}_2, 0) \right] \quad (38)$$

Evaluating this at our initial state gives us

$$\begin{aligned} \langle \mathcal{E}^\dagger(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \rangle &\approx \alpha^*(\mathbf{r}_1, 0) \alpha(\mathbf{r}_2, 0) \left\{ 1 - \frac{igz}{2k} \left[|\alpha(\mathbf{r}_2, 0)|^2 - |\alpha(\mathbf{r}_1, 0)|^2 \right] \right\} \\ &\approx \langle \mathcal{E}^\dagger(\mathbf{r}_1, z) \rangle \langle \mathcal{E}(\mathbf{r}_2, z) \rangle \end{aligned} \quad (39)$$

where the last approximate inequality comes from direct comparison with (31). In particular, this shows that, for any v_1 and v_2 , we have $H(\mathbf{r}_1, \mathbf{r}_2) \approx 0$.

We can also use this to write the expectation value of the number operator:

$$\langle n(v) \rangle = \langle a^\dagger(v) a(v) \rangle \approx |\langle a(v) \rangle|^2 \approx |\langle v | \alpha \rangle|^2 + \frac{gz}{k} \text{Im} \langle v | \alpha \rangle \langle \alpha, \alpha | \alpha, v \rangle \quad (40)$$

When both α and v are arbitrary Laguerre-Gaussian modes (any p, l, w), we have that $\text{Im} \langle \alpha | u \rangle \langle \alpha, \alpha | \alpha, v \rangle = 0$, so there is no first order correction to the occupation number.

The other two point field function is

$$\begin{aligned} \mathcal{E}(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) &\approx \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) + \mathcal{E}_0(\mathbf{r}_1, z) \delta \mathcal{E}(\mathbf{r}_2, z) + \delta \mathcal{E}(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) \\ &\approx \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) - \frac{igz}{2k} \left[\mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0^\dagger(\mathbf{r}_2, z) \mathcal{E}_0^2(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z) \mathcal{E}_0^2(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) \right] \\ &= \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) - \frac{igz}{2k} \left[\mathcal{E}_0^\dagger(\mathbf{r}_2, z) \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0^2(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z) \mathcal{E}_0^2(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) + \delta(\mathbf{r}_1 - \mathbf{r}_2) \mathcal{E}_0^2(\mathbf{r}_2, z) \right] \end{aligned} \quad (41)$$

The appearance of the delta function here is important, because now

$$\begin{aligned}\langle \mathcal{E}(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \rangle &\approx \alpha(\mathbf{r}_1, z) \alpha(\mathbf{r}_2, z) \left[1 - \frac{igz}{2k} \left[|\alpha(\mathbf{r}_2, z)|^2 + |\alpha(\mathbf{r}_1, z)|^2 \right] \right] - \frac{igz}{2k} \delta(\mathbf{r}_1 - \mathbf{r}_2) \alpha^2(\mathbf{r}_2, z) \\ &\approx \langle \mathcal{E}(\mathbf{r}_1, z) \rangle \langle \mathcal{E}(\mathbf{r}_2, z) \rangle - \frac{igz}{2k} \delta(\mathbf{r}_1 - \mathbf{r}_2) \alpha^2(\mathbf{r}_2, z)\end{aligned}\quad (42)$$

Therefore, we may conclude that

$$F(\mathbf{r}_1, \mathbf{r}_2) \approx -\frac{igz}{2k} \delta(\mathbf{r}_1 - \mathbf{r}_2) \alpha^2(\mathbf{r}_2, z), \quad (43)$$

$$f(v_1, v_2) \approx -\frac{igz}{2k} \langle v_1, v_2 | \alpha, \alpha \rangle. \quad (44)$$

It will be useful to put this result in a dimensionless form. First, let us assume the input mode has a well define beam spot size w , and a Rayleigh range $z_R = kw^2/2$. By defining

$$Z = \frac{gz \|\alpha\|^2}{4z_R} \quad (45)$$

and $u(\mathbf{r}) = w\alpha(\mathbf{r})/\|\alpha\|$ we get

$$f(v_1, v_2) \approx -iZ \langle v_1, v_2 | u, u \rangle \quad (46)$$

and

$$\Gamma(v_1, v_2) = \text{Re} \langle v_1 | v_2 \rangle + 2Z \text{Im} \langle v_1, v_2 | \alpha, \alpha \rangle \quad (47)$$

This leads us to a minimum quadrature variance for mode v of

$$\lambda_- = 1 - |2Z \langle v, v | u, u \rangle| \quad (48)$$

and an optimal Duan parameter for modes v_1 and v_2 of

$$D_{\text{opt}} = 1 - |2Z \langle v_1, v_2 | u, u \rangle|. \quad (49)$$

The corresponding angles are $\phi_{\text{squeezing}} = (-\pi/2 \pm \pi + \arg Z \langle v, v | \alpha, \alpha \rangle)/2$ and $\phi_{\text{duan}} = (-\pi/2 \pm \pi + \arg Z \langle v_1, v_2 | \alpha, \alpha \rangle)/2$

At this point, it becomes clear that the overlaps of the form $\langle v_1, v_2 | \alpha, \alpha \rangle$ are the ones that dominate this sort of process.

Consider the case where all the involved modes are vortices: $u(\mathbf{r}) = u_r(r) e^{il_u \phi}$ and similarly for the v functions. We have squeezing if $l_v = l_u$ and we have entanglement if $l_{v_1} + l_{v_2} = 2l_u$. If all the radial functions are positive, as is the case for $LG_{0,0}$ and $LG_{0,\pm 1}$, and α is positive, then the optimal angles are both $\text{sign}(g)\pi/4$.

D. Selection Rules for two-mode entanglement in the first order treatment

Consider the case where the input mode is $\alpha(\mathbf{r}) = \|\alpha\| LG_{0,l}(\mathbf{r})$, where

$$LG_{pl}(\mathbf{r}; w) = (-1)^p \frac{\mathcal{N}_{pl}}{w} \left(\frac{2r^2}{w^2} \right)^{|l|/2} L_p^{|l|} \left(\frac{2r^2}{w^2} \right) e^{-r^2/w^2 + il\phi} \quad (50)$$

is a Laguerre-Gaussian mode at the waist plane with radial order p , topological charge l and waist w . The normalization constant is

$$\mathcal{N}_{pl} = \sqrt{\frac{2}{\pi}} \sqrt{\frac{p!}{(p+|l|)!}} \quad (51)$$

Furthermore, let $v_2 \propto LG_{0,l_2}(\mathbf{r}; w_2)$ where w_2 is free and l_2 is only constrained by $l_2 \neq l$. Now, define $l_1 = 2l - l_2$ and $1/w_1^2 = 2/w^2 + 1/w_2^2$. We then have that

$$\begin{aligned} v_2^*(\mathbf{r})u^2(\mathbf{r}) &\propto \frac{\mathcal{N}_{0l_2}\mathcal{N}_{0l}^2}{w_2} \left(\frac{2r^2}{w_2^2}\right)^{|l_2|/2} \left(\frac{2r^2}{w^2}\right)^{|l|} e^{-r^2/w_1^2 + il_1\phi} \\ &= \frac{\mathcal{N}_{0l_2}\mathcal{N}_{0l}^2 w_1^{2|l|+|l_2|}}{w_2^{|l_2|+1} w^{2|l|}} \left(\frac{2r^2}{w_1^2}\right)^{|l_1|/2} \left(\frac{2r^2}{w_1^2}\right)^{p_{\max}} e^{-r^2/w_1^2 + il_1\phi} \end{aligned} \quad (52)$$

where we have defined

$$p_{\max} = \frac{1}{2}(2|l| + |l_2| - |2l - l_2|) = \begin{cases} 0, & ll_2 \leq 0 \\ \min(2|l|, |l_2|), & ll_2 > 0 \end{cases} \quad (53)$$

Then, by expanding the monomial that is raised to the power of $p_{\max}$ in terms of Laguerre polynomials, we get

$$v_2^*(\mathbf{r})u^2(\mathbf{r}) = \sum_{p=0}^{p_{\max}} C_p LG_{pl_1}(\mathbf{r}; w_1) \quad (54)$$

where

$$C_p = \frac{p_{\max}! \mathcal{N}_{0l_2} \mathcal{N}_{0l}^2 w_1^{2|l|+|l_2|+1}}{\mathcal{N}_{pl_1} w_2^{|l_2|+1} w^{2|l|}} \binom{p_{\max} + |l_1|}{p_{\max} - p} \quad (55)$$

Therefore, setting $v_1 = e^{i\phi} LG_{pl_1}(\mathbf{r}; w_1)$, which is orthogonal to v_2 , and considering the two mode subsystem formed by v_1, v_2 , we get an optimal Duan violation of

$$D_{\text{opt}} = \begin{cases} 1 - 2C_p |Z|, & p \leq p_{\max} \\ 1, & p > p_{\max} \end{cases} \quad (56)$$

Therefore, when $p \leq p_{\max}$, we have bipartite entanglement in this two-mode subsystem.

In Fig. IIID we show the violation of the Duan criterion as a function of Z for a few different combinations of l_0 and l_2 . The selection rules can be clearly observed here: in all cases, the modes with $p > p_{\max}$ fail to violate the entanglement criterion (which does not mean absence of entanglement).

E. Multipartite entanglement

The paper [2] introduces a generalized class of entanglement witnesses $\mathcal{Z}$ such that $\text{Tr } \mathcal{Z}\gamma < 1$ implies that the state ρ corresponding to a correlation matrix γ is entangled (in a sense made more precise later), while $\text{Tr } \mathcal{Z}\gamma > 1$ for some separable state. The Duan criterion is a special case of a witness $\mathcal{Z}$ such that

$$\mathcal{Z} = \frac{1}{4} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}. \quad (57)$$

Multipartite entanglement is a subtle topic, because separability becomes more complicated once there are more than two parties. To handle this, [2] introduces two witnesses types:

- Witness for entanglement with respect to a given partition. Here, one fixes a partition and asks whether the state is separable according to this partition. There is a numerical routine called FullyWit that finds the optimal $\mathcal{Z}$ for a given state, in the sense that the violation $1 - \text{Tr } \mathcal{Z}\gamma$ is maximized.
- Witness for genuine multipartite entanglement. Suppose that now we have many parties and we want to decide if the state is separable with respect to these parties. One could then check individually all of the bipartitions with the previous method. For example, if we have three parties A, B and C , one could check individually the bipartitions $AB|C$, $AC|B$ and $A|BC$. This is actually sufficient for pure states, but, for mixed ones, we could write a state as a mixture of biseparable ones, say $\rho = \frac{1}{3}\rho_{AB|C} + \frac{1}{3}\rho_{AC|B} + \frac{1}{3}\rho_{A|BC}$. The full state ρ is not separable with respect to any bipartition, but its individual constituents are. We then say that a state is genuine multipartite entangled if it cannot be written as a convex sum of bipartite separable states. The reference [2] also introduces a routine, MultiWit, that finds an optimized witness for genuine multipartite entanglement.

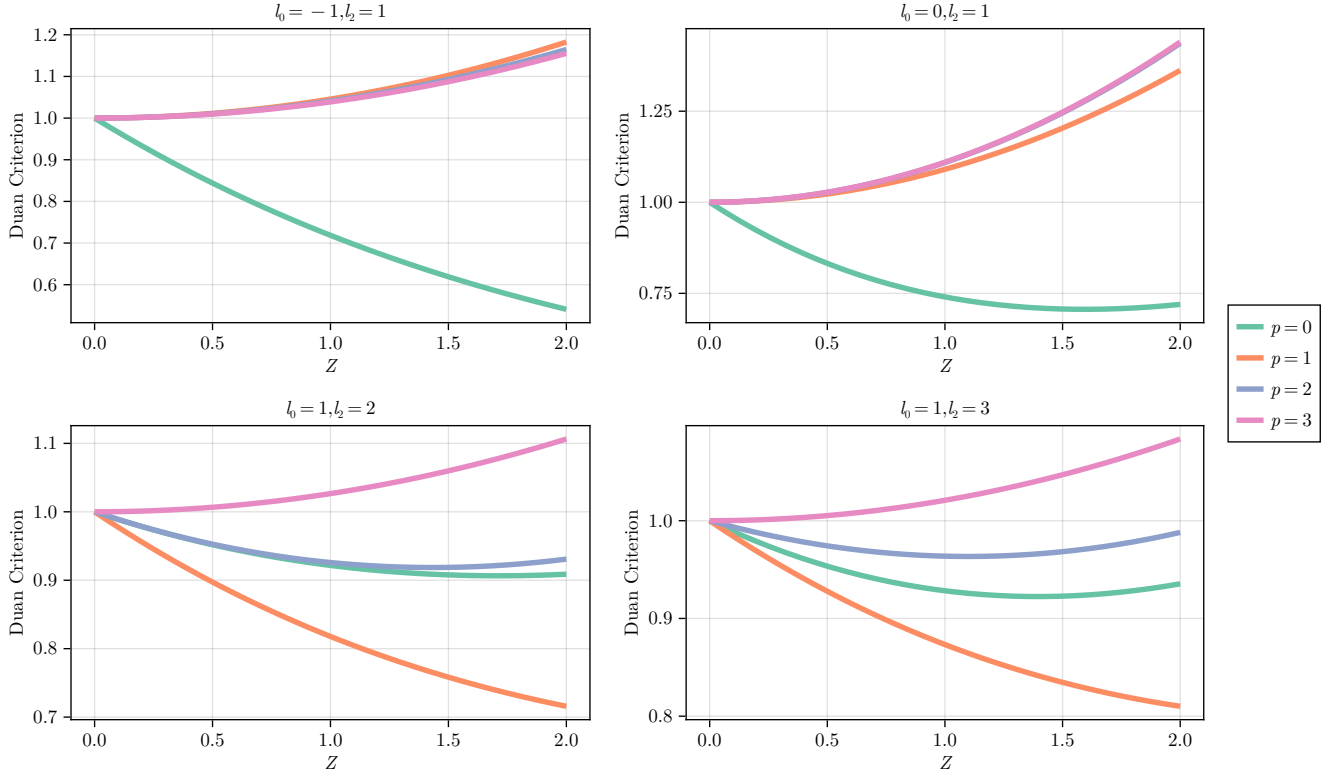


FIG. 1. Duan Criterion Violation

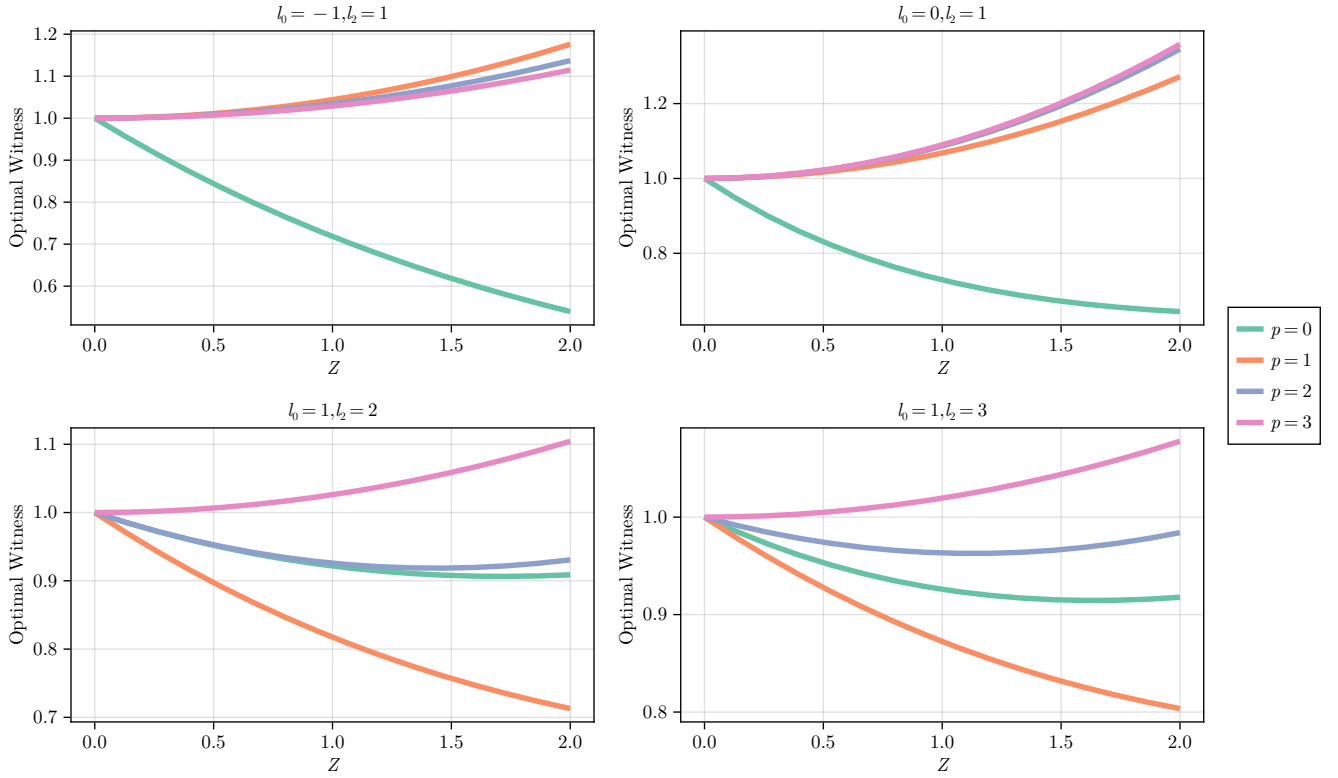


FIG. 2. Violation of the optimized witness in the bipartite case.

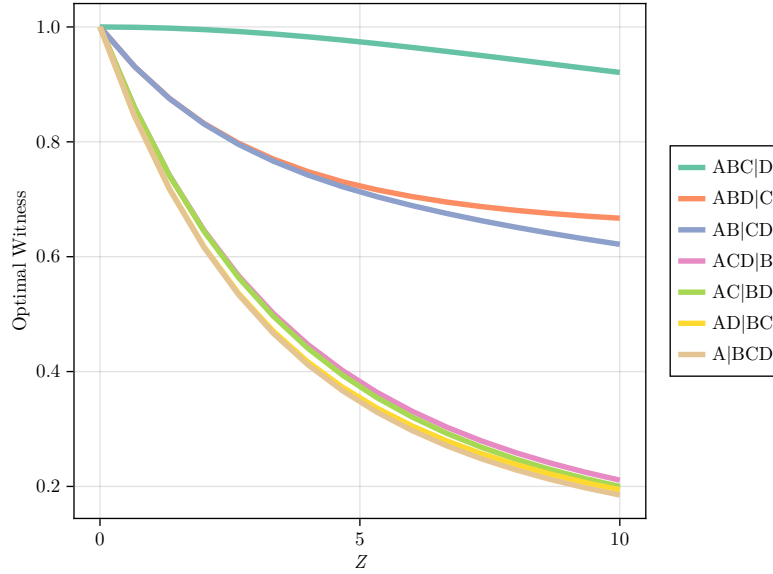


FIG. 3. Violation of the optimized witness for bipartite entanglement over multiple modes.

Now we employ these types of optimal entanglement witnesses for our Kerr system.

In Fig. IIIE we consider the same setup as in Fig. IIID, but use the optimized witness. We do not see a large difference with respect to IIID, which shows that the optimization over the angles used to arrive at (19) already carries us a long way.

Now we turn to more parties. We set $l_0 = 1$ and $l_2 = 2$, corresponding to the bottom left plot of figures IIID and IIIE. Furthermore, we define the parties as $A = LG_{0,2}(w_2 = w)$, $B = LG_{0,-1}(w_1)$, $C = LG_{1,-1}(w_1)$ and $D = LG_{2,-1}(w_1)$. According to the previous results, we only had entanglement between $A|B$ and $A|C$, while there were inconclusive results when considering $A|C$. In Fig. IIIE we use the FullyWit routine to find optimized witnesses for the all the bipartitions of these four modes. The lines seem to cluster in three groups

- $ACD|B$, $AC|BD$, $AD|BC$ and $A|BCD$ show the highest violation. These share the fact that A and B belong to different parties, suggesting that these share, in some sense, more entanglement.
- $ABD|C$ and $AB|CD$ show an intermediate violation, still maintaining a linear start. Here, A and B belong to the same party, while C is in the other one.
- $ABC|D$ shows the smallest violation, starting with a quadratic fall. Here, all the modes that were bipartite entangled sit on the same party.

Finally, we turn to the genuine multipartite case. In Fig. IIIE, we show the witness for sets of the form $\{LG_{0,1}(w_2 = w)\} \cup \{LG_{p,3}(w_2 = w) \mid p \leq P\}$, with P varying from 0 to 3, while maintaining $l_0 = 2$. We once again see that the selection rule derived in the two mode case holds: we see a much stronger violation for the sets with $P \leq p_{\max} = 1$. However, although the $P = 0$ case seems linear, $P = 1$ seems more quadratic, although this is just visual.

IV. CONCLUSION

Appendix A: Purely nonlinear approach

Let us make a modification to our equations of motion where

$$2ik\partial_z\mathcal{E}(\mathbf{r}, z) = \int_{\mathbb{R}^2} g(\mathbf{s})\mathcal{E}^\dagger(\mathbf{r} - \mathbf{s}, z)\mathcal{E}(\mathbf{r} - \mathbf{s}, z)ds \mathcal{E}(\mathbf{r}, z). \quad (\text{A1})$$

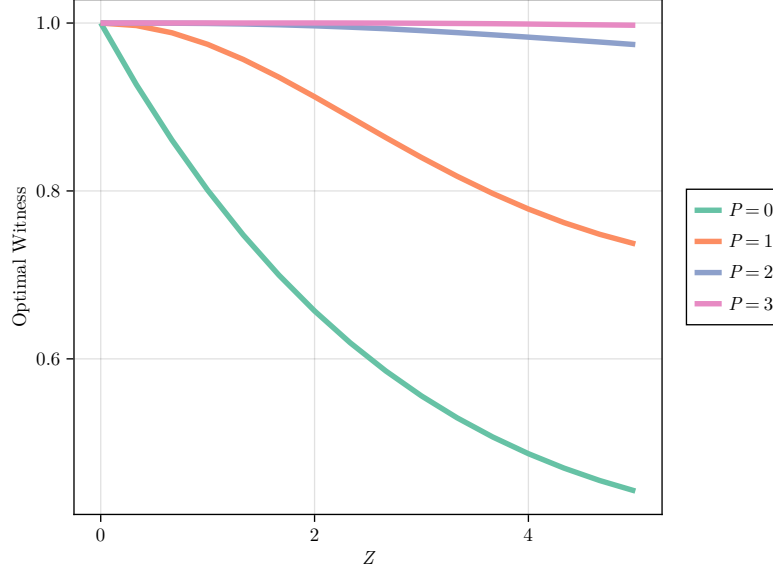


FIG. 4. Violation of the optimized witness for genuine multipartite entanglement.

In this form, the equation has a constant of motion $\mathcal{E}^\dagger(\mathbf{r}, z)\mathcal{E}(\mathbf{r}, z)$, which leads us to the exact solution

$$\begin{aligned}\mathcal{E}(\mathbf{r}, z) &= \exp \left[-\frac{iz}{2k} \int_{\mathbb{R}^2} g(\mathbf{s}) \mathcal{E}^\dagger(\mathbf{r} - \mathbf{s}, 0) \mathcal{E}(\mathbf{r} - \mathbf{s}, 0) d\mathbf{s} \right] \mathcal{E}(\mathbf{r}, 0) \\ &=: \exp \left[\int_{\mathbb{R}^2} \left(e^{-izg(\mathbf{s})/2k} - 1 \right) \mathcal{E}^\dagger(\mathbf{r} - \mathbf{s}, 0) \mathcal{E}(\mathbf{r} - \mathbf{s}, 0) d\mathbf{s} \right] : \mathcal{E}(\mathbf{r}, 0)\end{aligned}\quad (\text{A2})$$

In the last line, we used an important result known as the normal ordering theorem [3, 4]. This allows us to simply calculate the expectation value for the field in an initial coherent state:

$$\langle \mathcal{E}(\mathbf{r}, z) \rangle = \exp \left[\int_{\mathbb{R}^2} \left(e^{-izg(\mathbf{s})/2k} - 1 \right) |\alpha(\mathbf{r} - \mathbf{s}, 0)|^2 d\mathbf{s} \right] u(\mathbf{r}, 0) \quad (\text{A3})$$

We see that (31) is then a first order approximation of (A3) with $g(\mathbf{s}) = g_0\delta(\mathbf{s})$. Nonetheless, we see that we get singularities in the full expression when using this form of g . Let us then choose g as proportional to an approximation of the delta function. For definiteness, let us define

$$\delta_0(x, a) = \frac{\Theta(a/2 - |x|)}{a}, \quad (\text{A4})$$

which approximates the delta distribution as $a \rightarrow 0$. Here, $\Theta(x)$ is the step function. Let us denote $\delta_0(\mathbf{r}, a) = \delta_0(x, a)\delta_0(y, a)$. We then choose $g(\mathbf{s}) = g_0\delta_0(\mathbf{s})$.

We will usually make the approximation that the support of the response function g is much smaller than the characteristic distances of the mode α . We can then make the approximation

$$\langle \mathcal{E}(\mathbf{r}, z) \rangle = \exp \left[G(z) |\alpha(\mathbf{r}, 0)|^2 \right] \alpha(\mathbf{r}, 0) \quad (\text{A5})$$

where

$$G(z) = \int_{\mathbb{R}^2} \left(e^{-izg(\mathbf{s})/2k} - 1 \right) d\mathbf{s} = a^2 \left(e^{-izg_0/2ka^2} - 1 \right) \quad (\text{A6})$$

Notice then that, in the general case, this is quite dependent on the value of a . Nonetheless, when $z|g_0|/2ka^2 \ll 1$, we get

$$G(z) \approx -\frac{izg_0}{2k}, \quad (\text{A7})$$

which is independent of a .

Following the result of [4] we can calculate

$$F(\mathbf{r}_1, \mathbf{r}_2) \approx \alpha(\mathbf{r}_1, 0)\alpha(\mathbf{r}_2, 0) \exp \left[G(z)(|\alpha(\mathbf{r}_1)|^2 + |\alpha(\mathbf{r}_2)|^2) \right] \times \left[e^{-ig(\mathbf{r}_1 - \mathbf{r}_2)z/2k + |\alpha(\mathbf{r}_1)|^2 G_1(\mathbf{r}_1 - \mathbf{r}_2)} - 1 \right] \quad (\text{A8})$$

where

$$\begin{aligned} G_1(\mathbf{r}, z) &= \frac{1}{2} \int_{\mathbb{R}^2} (e^{-izg(\mathbf{s} - \mathbf{r})} - 1)(e^{-izg(\mathbf{s})} - 1) d\mathbf{s} \\ &= \frac{a^4 (e^{-ig_0 z/2ka^2} - 1)^2}{2} \delta_1(\mathbf{r}, a) \\ &\approx -\frac{(g_0 z/2k)^2}{2} \delta_1(\mathbf{r}, a) \end{aligned} \quad (\text{A9})$$

where we have defined

$$\delta_1(x, a) = \frac{(a - |x|)\Theta(a - |x|)}{a^2} \quad (\text{A10})$$

as another approximation of the delta function.

Notice, then, that, for modes with variation on a scale much larger than a , the second line of (A8) is essentially proportional to $\delta(\mathbf{r}_1 - \mathbf{r}_2)$, and we may write

$$F(\mathbf{r}_1, \mathbf{r}_2) \approx \alpha(\mathbf{r}_1, z)^2 \bar{F}(\mathbf{r}_1, a) \delta(\mathbf{r}_1 - \mathbf{r}_2) \quad (\text{A11})$$

where

$$\alpha(\mathbf{r}, z) = \alpha(\mathbf{r}, 0) e^{-ig_0 z |\alpha(\mathbf{r}, 0)|^2 / 2k} = \alpha(\mathbf{r}, 0) e^{-iZ |u(\mathbf{r}, 0)|^2 / 4} \quad (\text{A12})$$

is the evolved field under the classical Kerr interaction and

$$\bar{F}(\mathbf{r}, a) = \int_{\mathbb{R}^2} \left\{ \exp \left[-i \frac{g_0 z}{2k} \delta_0(\mathbf{s}, a) - \frac{1}{2} \left(\frac{g_0 z}{2k} \right)^2 \delta_1(\mathbf{s}, a) |\alpha(\mathbf{r})|^2 \right] - 1 \right\} d\mathbf{s} \quad (\text{A13})$$

In general, this function presents a strong dependency on a . Nonetheless, if we also assume that $z^2 |g_0|^2 |\alpha(\mathbf{r})|^2 / 4k^2 a^2 \ll 1$, then we may expand the exponential in a Taylor series and we get

$$\bar{F}(\mathbf{r}, a) = -\frac{g_0 z}{2k} \left[i + \frac{g_0 |\alpha(\mathbf{r})|^2 z}{2k} \right], \quad (\text{A14})$$

which does not depend on a . Substituting this result, we obtain

$$\begin{aligned} F(\mathbf{r}_1, \mathbf{r}_2) &\approx -\frac{g_0 z}{2k} \left[i + \frac{g_0 |\alpha(\mathbf{r})|^2 z}{2k} \right] \alpha(\mathbf{r}_1, z)^2 \delta(\mathbf{r}_1 - \mathbf{r}_2) \\ &= -Z \left[i + Z |u(\mathbf{r})|^2 \right] u(\mathbf{r}_1, z)^2 \delta(\mathbf{r}_1 - \mathbf{r}_2) \end{aligned} \quad (\text{A15})$$

and

$$f(v_1, v_2) = -iZ \langle v_1, v_2 | u(z), u(z) \rangle - Z^2 \langle v_1, v_2, u(z) | u(z), u(z), u(z) \rangle \quad (\text{A16})$$

Once again, following [4], we calculate

$$H(\mathbf{r}_1, \mathbf{r}_2) = \alpha^*(\mathbf{r}_1, z) \alpha(\mathbf{r}_2, z) (e^{|\alpha(\mathbf{r}_1)|^2 G_2(\mathbf{r}_1 - \mathbf{r}_2)} - 1) \quad (\text{A17})$$

where

$$G_2(\mathbf{r}) = a^4 \left| e^{ig_0 z/2ka^2} - 1 \right|^2 \delta_1(\mathbf{r}, a) \approx \left(\frac{g_0 z}{2k} \right)^2 \delta_1(\mathbf{r}, a) \quad (\text{A18})$$

Therefore,

$$\begin{aligned} H(\mathbf{r}_1, \mathbf{r}_2) &\approx \left(\frac{g_0 z}{2k}\right)^2 |\alpha(\mathbf{r}_1, z)|^4 \delta(\mathbf{r}_1 - \mathbf{r}_2) \\ &= Z^2 |u(\mathbf{r}_1, z)|^4 \delta(\mathbf{r}_1 - \mathbf{r}_2) \end{aligned} \quad (\text{A19})$$

and

$$h(v_1, v_2) = Z^2 \langle v_1, u(z), u(z) | v_2, u(z), u(z) \rangle \quad (\text{A20})$$

Notice that $h(v, v) \geq 0$, so it reduces both squeezing and the violation of Duan's inequality.

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- [1] L.-M. Duan, G. Giedke, J. I. Cirac, and P. Zoller, Inseparability criterion for continuous variable systems, Physical review letters **84**, 2722 (2000).
 - [2] P. Hyllus and J. Eisert, Optimal entanglement witnesses for continuous-variable systems, New Journal of Physics **8**, 51 (2006).
 - [3] K. J. Blow, R. Loudon, S. J. D. Phoenix, and T. J. Shepherd, Continuum fields in quantum optics, Phys. Rev. A **42**, 4102 (1990).
 - [4] K. J. Blow, R. Loudon, and S. J. D. Phoenix, Exact solution for quantum self-phase modulation, J. Opt. Soc. Am. B **8**, 1750 (1991).